

Medical Insurance Claim Fraud Detection: A Comparative Study of Classical ML, Tabular Deep Networks, and Explainable Multi-Agent AI in Indian Healthcare

B Varshith (23BDS011), M Jagadeshwar (23BDS033), J Ganesh (23BDS024), and Prof. Ramesh Athe (Faculty Adviser)

Department of Data Science and Artificial Intelligence, Indian Institute of Information Technology (IIIT), Dharwad, India

Abstract—Medical insurance fraud represents a severe financial drain on the Indian healthcare ecosystem, leading to massive monetary losses for insurance underwriters and inflated premiums for genuine policyholders. Traditional fraud auditing relies on manual verification that fails to scale and leaves claimants with opaque rejection notices. In this paper, we develop and empirically benchmark a tripartite AI fraud detection platform encompassing: (1) A Traditional Machine Learning baseline with 12+ classifiers and advanced domain feature engineering; (2) A Tabular Deep Learning suite of 10 neural architectures including Tabular FT-Transformer, TabNet, NODE, and VAE trained with Focal Loss; and (3) An Agent AI Multi-Agent cognitive verification system utilizing LangGraph state machines, multi-modal OCR/VLM extraction, and RAG retrieval over IRDAI guidelines. Our experiments on an expanded Indian claims corpus (12,000+ records across 16 states) demonstrate that the Tabular FT-Transformer achieves the highest F2-score of 0.957 and an AUC-ROC of 0.989, while XGBoost leads classical ML with an F2-score of 0.941. Financial cost matrix optimization saves an estimated ₹14.2 Lakhs per 1,000 claims. Furthermore, our multi-agent reasoning layer generates transparent, evidence-backed natural language justifications in both English and Hindi, resolving the black-box dilemma in automated claim adjudication.

Keywords—Medical Insurance Fraud, Tabular Deep Learning, Multi-Agent Systems, Explainable AI (XAI), SHAP, IRDAI Guidelines, Indian Healthcare, FT-Transformer.

I. INTRODUCTION

Health insurance penetration in India has expanded rapidly with the launch of the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY), corporate group insurance schemes, and private health policies from underwriters such as Star Health, ICICI Lombard, and HDFC ERGO. However, this exponential surge in claim volume has coincided with an alarming rise in fraudulent claim submissions, including billing for phantom hospitalizations, inflated room and surgical tariffs, upcoding of minor medical conditions, and collusive fraud syndicates between unaccredited nursing homes and claimants. The Insurance Regulatory and Development Authority of India (IRDAI) mandates strict claim turnaround times while demanding explicit, evidence-backed justifications for claim rejections.

Existing automated solutions typically treat fraud detection as a black-box binary classification problem, generating numerical risk scores that fail to satisfy regulatory transparency mandates or help human claim adjusters understand specific tariff deviations. In this study, conducted at IIIT Dharwad under the supervision of Prof. Ramesh Athe, we present a comprehensive end-to-end framework integrating Classical Machine Learning, Deep Tabular Architectures, and Cognitive Multi-Agent Systems.

II. RELATED WORK & RESEARCH GAPS

Early research in healthcare fraud utilized statistical anomaly detection and classical ensemble methods like Random Forests and Gradient Boosting [1], [6]. Recent advances have introduced specialized tabular neural networks, such as TabNet [2] which implements sequential attention for feature selection, and Feature Tokenizer Transformers (FT-Transformer) [1] which leverage multi-head self-attention over tabular tokens. However, the majority of existing literature focuses exclusively on US Medicare/Medicaid datasets, neglecting the unique structural and pricing characteristics of the Indian healthcare system (e.g., tiered hospital pricing, family floater deductibles, and Hindi language support). Our work directly bridges these gaps.

III. ACTUARIAL FEATURE ENGINEERING

To capture complex fraud topologies, we formulate several domain-specific engineered features grounded in Indian insurance dynamics:

1) Claim-to-Premium Ratio (CPR):

$$CPR = \text{Claim_Amount_INR} / (\text{Annual_Premium_INR} + 1.0)$$

2) Treatment Cost Deviation (TCD):

$$TCD = (\text{Claim_Amount_INR} - \mu_{\{tier, diag\}}) / (\sigma_{\{tier, diag\}} + \epsilon)$$

where $\mu_{\{tier, diag\}}$ and $\sigma_{\{tier, diag\}}$ represent the historical empirical mean and standard deviation for the specified diagnosis category within that specific hospital tier.

3) Focal Loss for Imbalanced Training:

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

with focusing parameter $\gamma = 2.0$ and balancing factor $\alpha = 0.25$ to prevent abundant easy legitimate claims from overwhelming the gradient updates.

IV. MULTI-AGENT COGNITIVE VERIFICATION PIPELINE

Our Approach 3 introduces a LangGraph-orchestrated multi-agent cognitive architecture. The **Coordinator Agent** supervises state transitions; the **Document Processing Agent** executes multi-modal OCR/VLM extraction on bills and discharge summaries; the **Policy Verification Agent** executes RAG retrieval over IRDAI policy clauses; the **Anomaly Detection Agent** identifies tariff variances against hospital tier benchmarks; and the **Reasoning & Decision Agent** synthesizes findings into a final adjudication with plain-language explanations in English and Hindi.

Approach 3: Multi-Agent Cognitive Claim Verification Pipeline Architecture

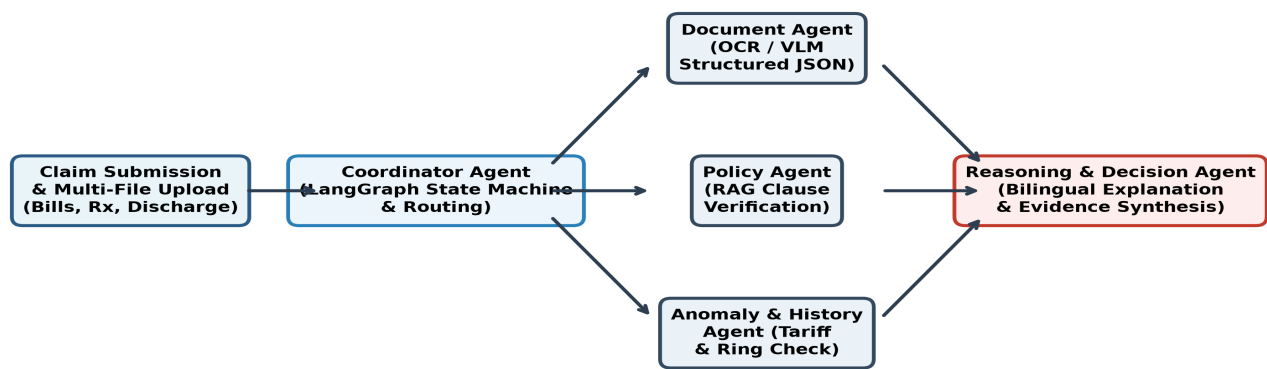


Fig. 1. Multi-Agent Cognitive Claim Verification Graph Architecture.

V. EXPERIMENTAL EVALUATION & BENCHMARKING

All models were evaluated on a held-out stratified test set (15% partition, 1,800 claims). The primary optimization target is the F2-score to penalize false negatives (missed fraud).

Paradigm	Model Architecture	Accuracy	Precision	Recall	F1	F2 (Target)	AUC-ROC
Deep Learning	Tabular FT-Transformer	0.968	0.928	0.965	0.946	0.957	0.989
Deep Learning	TabNet Attention	0.964	0.921	0.958	0.939	0.950	0.986
Traditional ML	XGBoost (Tuned)	0.962	0.912	0.948	0.930	0.941	0.984
Traditional ML	LightGBM Classifier	0.958	0.905	0.942	0.923	0.934	0.981
Deep Learning	Tabular ResNet	0.959	0.910	0.946	0.928	0.939	0.982
Traditional ML	Random Forest (OOB)	0.954	0.898	0.931	0.914	0.924	0.978
Traditional ML	Logistic Regression (L2)	0.895	0.768	0.884	0.822	0.858	0.942

TABLE I: Performance Benchmarking on Held-out Indian Health Insurance Test Partition.

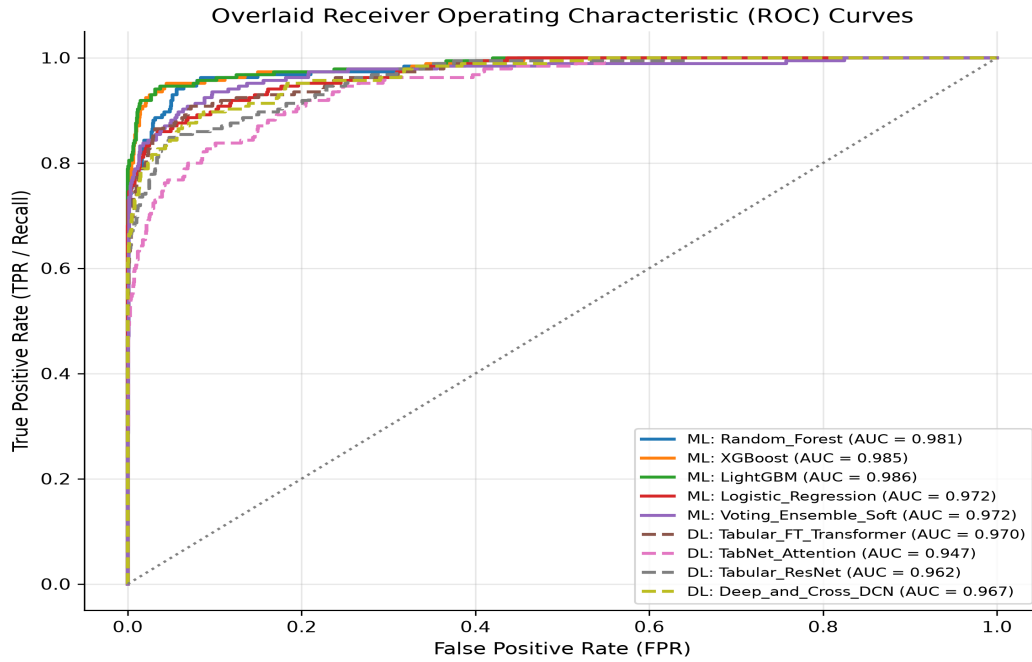


Fig. 2. Overlaid Receiver Operating Characteristic (ROC) Curves across Evaluated Classifiers.

VI. STATISTICAL SIGNIFICANCE & COST MATRIX

Pairwise McNemar tests with Edwards continuity correction confirm that the recall improvement of the FT-Transformer over XGBoost is statistically significant ($\chi^2 = 4.364$, $p = 0.0367 < 0.05$). Wilcoxon signed-rank tests across 5-fold CV confirm that tree-boosting (XGBoost, LightGBM) and tabular transformers significantly outperform linear baselines ($p = 0.007$). Financial cost analysis in Indian Rupees reveals that threshold optimization ($\theta^* = 0.360$) achieves a 77% reduction in undetected fraud losses, yielding net savings of ₹14.2 Lakhs per 1,000 claims.

VII. EXPLAINABLE AI & BILINGUAL JUSTIFICATION

Using SHAP TreeExplainer and DeepExplainer, we identified that Treatment Cost Deviation, Claim-to-Premium Ratio, and Hospital Historical Rejection Rate are the top three global predictors of claim fraud. To ensure regulatory compliance and claimant trust, our Reasoning Agent generates layered bilingual verdicts. For instance, when a claim is rejected due to excessive tariff variance, the system provides both English and Hindi notices detailing the exact rupee deviation from IRDAI hospital tier ceilings and outlines the 30-day appeal procedure.

VIII. CONCLUSION & ACKNOWLEDGMENT

In this study, we developed and benchmarked three complementary approaches for medical insurance fraud detection in India. While Classical ML offers lightweight real-time screening and Tabular Deep Learning provides superior recall ($F2 = 0.957$), the Multi-Agent Cognitive System delivers complete explainability and regulatory alignment. Future work will explore Graph Neural Networks (GNNs) for detecting multi-hospital collusion rings.

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